

Data Warehousing with Hive

Slowly Changing Dimensions (SCDs): Handling Changes Over Time

Problem statement

Existing Solution

Why Hive in Data Warehousing?

Understanding Hive’s Architecture

What are different HQL commands that are used commonly?

Hive Execution Engine

Where Should You Use Apache Hive?

PK_Asin	BookName	PAuthorId	SAuthorId	Language	PaperBack	ItemWeight	UnitPrice	Currency
345803787	Crazy Rich Asians	1001	1780	English	544	372	999	INR
1760290785	China Rich Girlfriend	1038	6701	English	400	279	285	INR
1786091089	Rich People Problems	2479	5310	English	560	383	399	INR

slowly changing Dimension-? Handle changes over time

SCD Type 0 :- Do Nothing Approach
↳ No changes are captured



SCD Type 1 :- Overwriting the old value
↳ Simplicity
↳ history is not maintained.

* SCD Type 2 :- Keeping a full history

↳ Full history

↳ Complexity ✓

↳ Storage

↳ PK violation ✓

Surrogate keys:-

↳ is an artificial system identifier

↳ Use 2 to differentiate b/w records in a table.

types =

- On level of history
- less storage or violation
- No tracking

SCD Type 4 :- Using a historical table
 one table :- Current values
 Second table :- historical values

- ⇒ More Storage
- ⇒ More Complexity (Jaws)
- ⇒ Duplicates

* SCD Type 6 $\Rightarrow$ the Hybrid Approach
SCD (V + SCD 2T)

SCD 3

→ Comprehensive tracking:-

→ Complexity

→ Storage

Design über Datenstrukturen

① Key Entities:-

Rides

Drivers

Passengers

Payments

Reviews

Date

Location

Support

Rating

Fare

②

Key Facts:-

Ride-details

Ridid

Daarid

Cid

Vid

IK

kar

51

Pickuploc ID	
Drop off location ID	
Timeid	
Distance	
Fare	
Duration	

Facts

Dimension tables

① Driver

Driverid (PK)
name
age
Gender
License
Rating
experience

② Customer

Cid PK
name
age
Ph-no
Cust-loc
email
gender
Clot-rating
Ph-type

③ Location ()
()
Time

2 (4) time
 (5) vehicle
 ()
 SCD
 fact - index

order_id	C_id	driver_id	Pickup loc-id	Drop loc-id	order status	fare	Surge	timestamp
101	1001	201	2001	3001	C	1.0	1.2	10-20-2019

→ ① Customer Dimension
 8CD-2
 ② Driver Dimension
 ③ Location Dimension
 ④ Pricing dimension

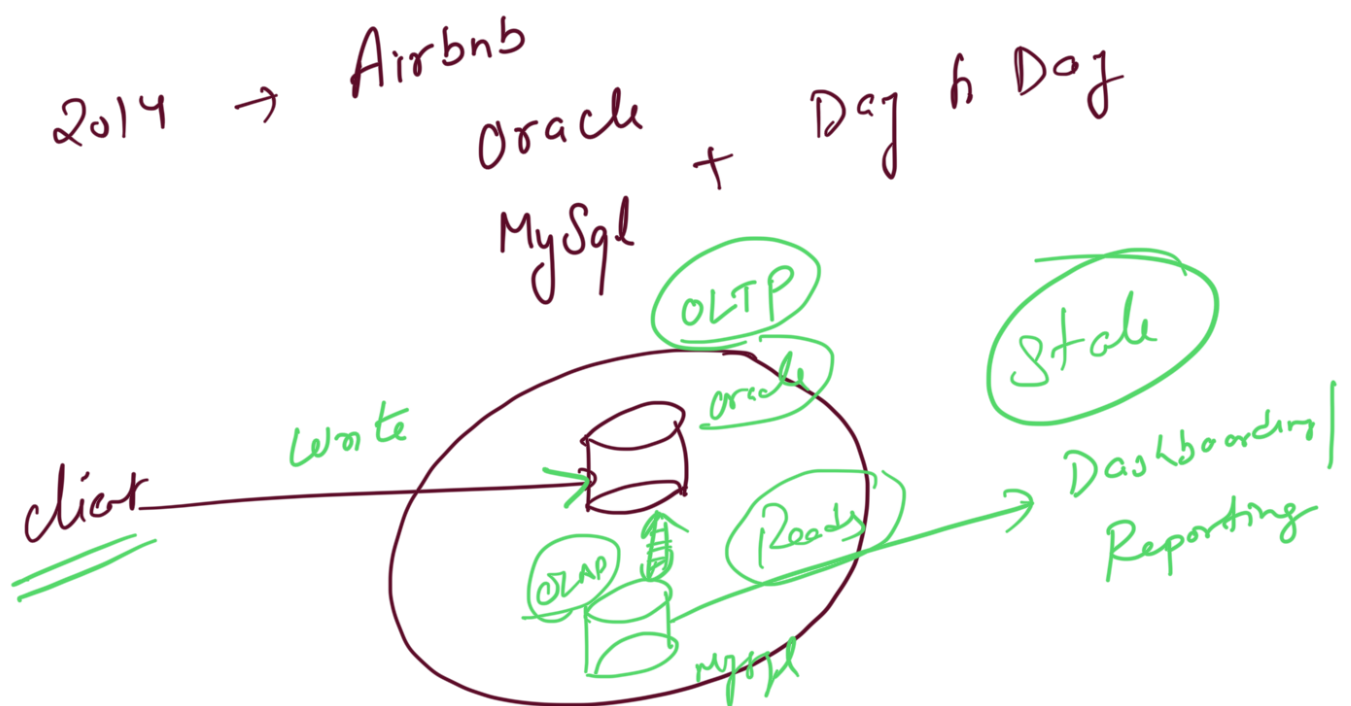
```

CREATE TABLE dim_customer (
  customer_id INT,
  customer_name VARCHAR(255),
  email VARCHAR(255),
  phone VARCHAR(20),
  status VARCHAR(50),
  start_date DATE,
  end_date DATE,
  active_flag CHAR(1), -- 'Y' for active, 'N' for inactive
  PRIMARY KEY (customer_id, start_date)
);
  
```

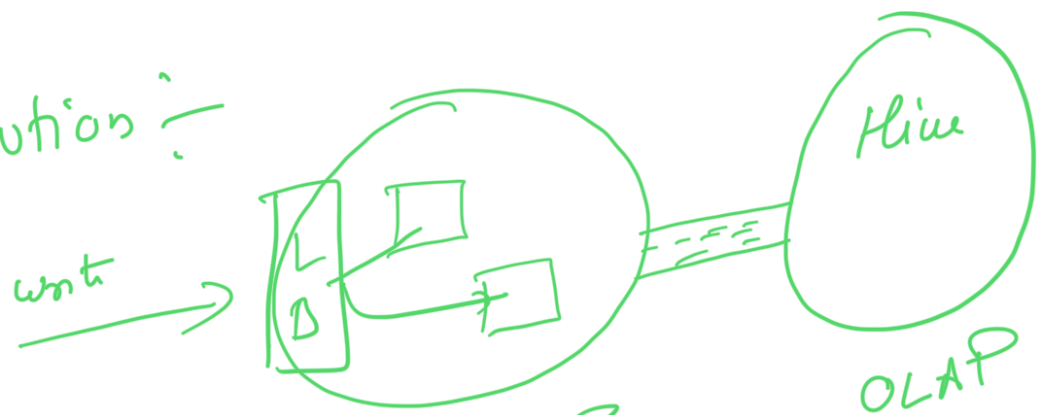
Surrogate key but PK not allowed

```
CREATE TABLE dim_driver (
    driver_id INT,
    driver_name VARCHAR(255),
    rating DECIMAL(3,2),
    car_type VARCHAR(50),
    prev_car_type VARCHAR(50), -- SCD Type 3
    start_date DATE,
    end_date DATE,
    active_flag CHAR(1),
    history_flag CHAR(1), -- 'Y' for historical records
    PRIMARY KEY (driver_id, start_date)
);
```

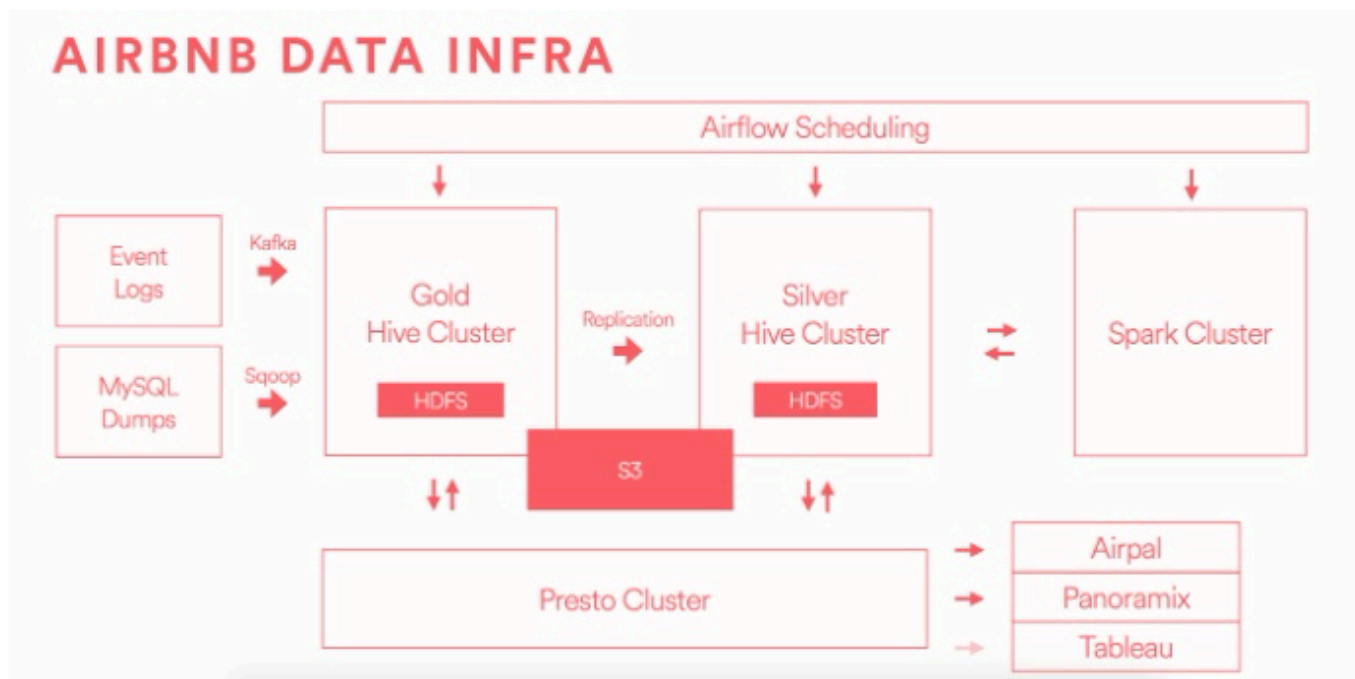
Problem statement



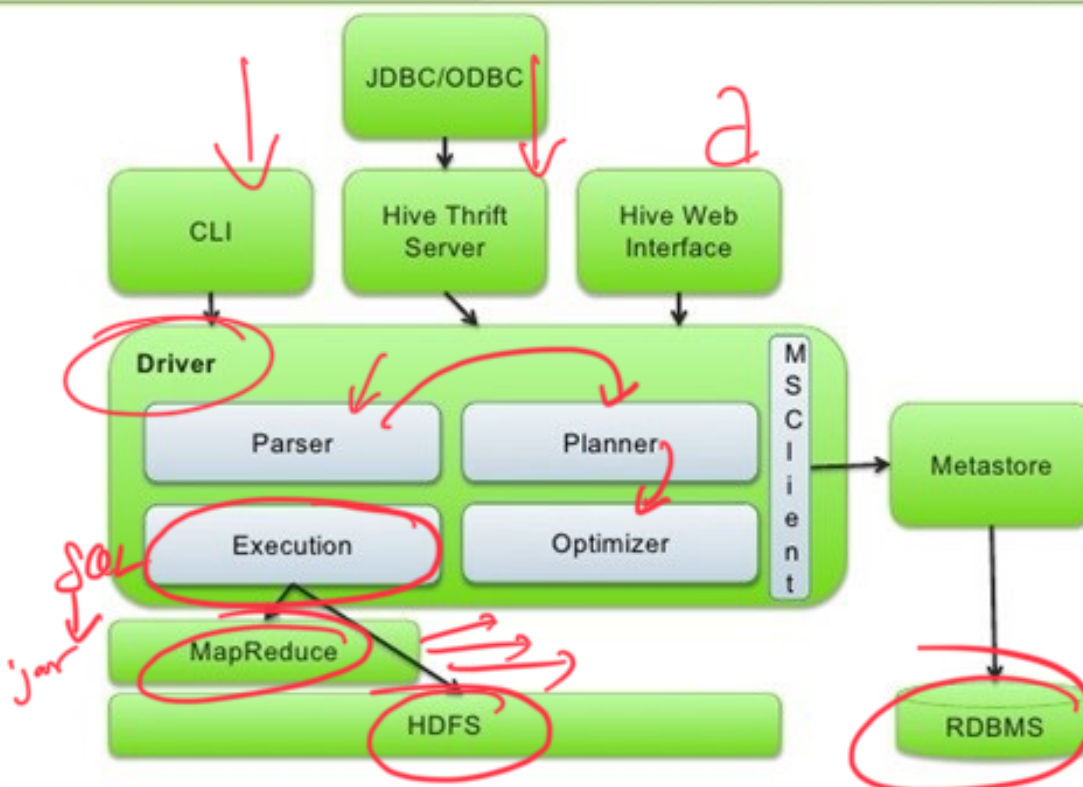
Solution :-



RDBMS OLTP
Hive is an open source Datawarehouse
which can store infinite no of records
and can process them in a fast manner.



→ Distributed, fault tolerant Dwh System
that enables analytics at a massive
scale.



① Hadoop core

HDFS = storage
MapReduce = Processing

SQL
↓
Java
↓
class
↓
Jar

② Meta store → in memory DB

(Derby)

↳ mysql

↳ metadata of each table

Col names
col datatype

③ Driver